



High-Performance Computing
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Introduction Simple Gaussian Blur

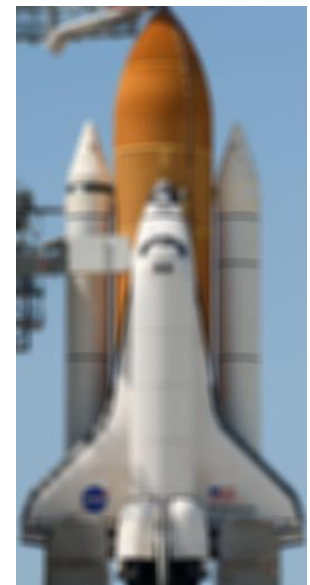
Simple Gaussian Blur

- Image processing technique to reduce noise and soften details
- Applies a Gaussian function to smooth pixel values
- Apply Gaussian function to each pixel and create a weighted average of all surrounding pixel values in input image
- Result of this weighted average is then written to the corresponding pixel in the output image

Input



Output



Gaussian Function Example

- 2D Gaussian filter kernel (Size 3, Sigma 1.0)



Input

0.0751	0.1238	0.0751
0.1238	0.2041	0.1238
0.0751	0.1238	0.0751



Output